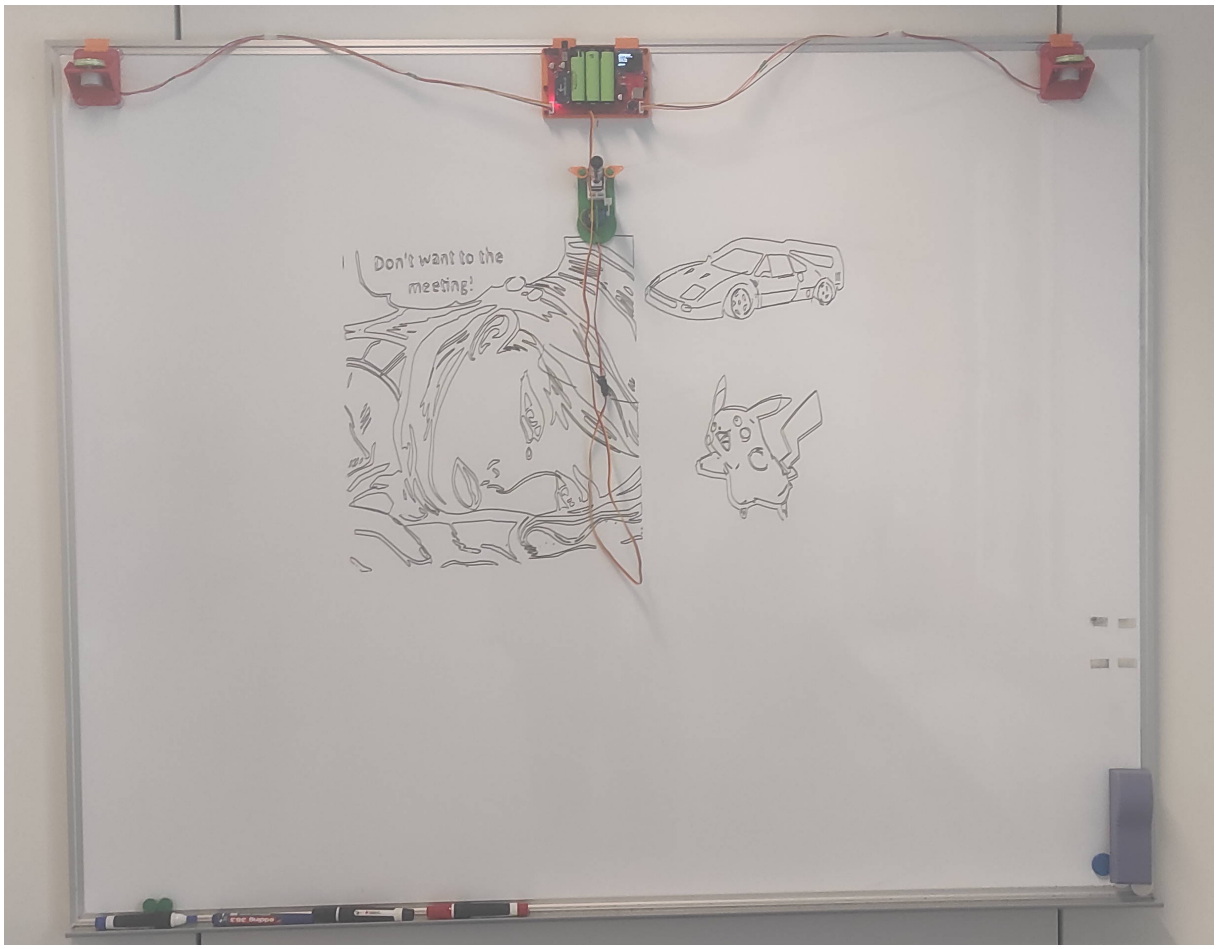
Wall plotter / polargraph with UNO or ESP32

for whiteboards

By Thorsten Hartwig, August2022



The idea of a wall plotter that moves a pen on two strings across a surface is now somewhat older.Such projects are known under names like Makelangelo, Polargraph, Wall Drawer and Wall Plotter. A little googling will reveal many projects. Watch the videos in the project, there's not much to it.

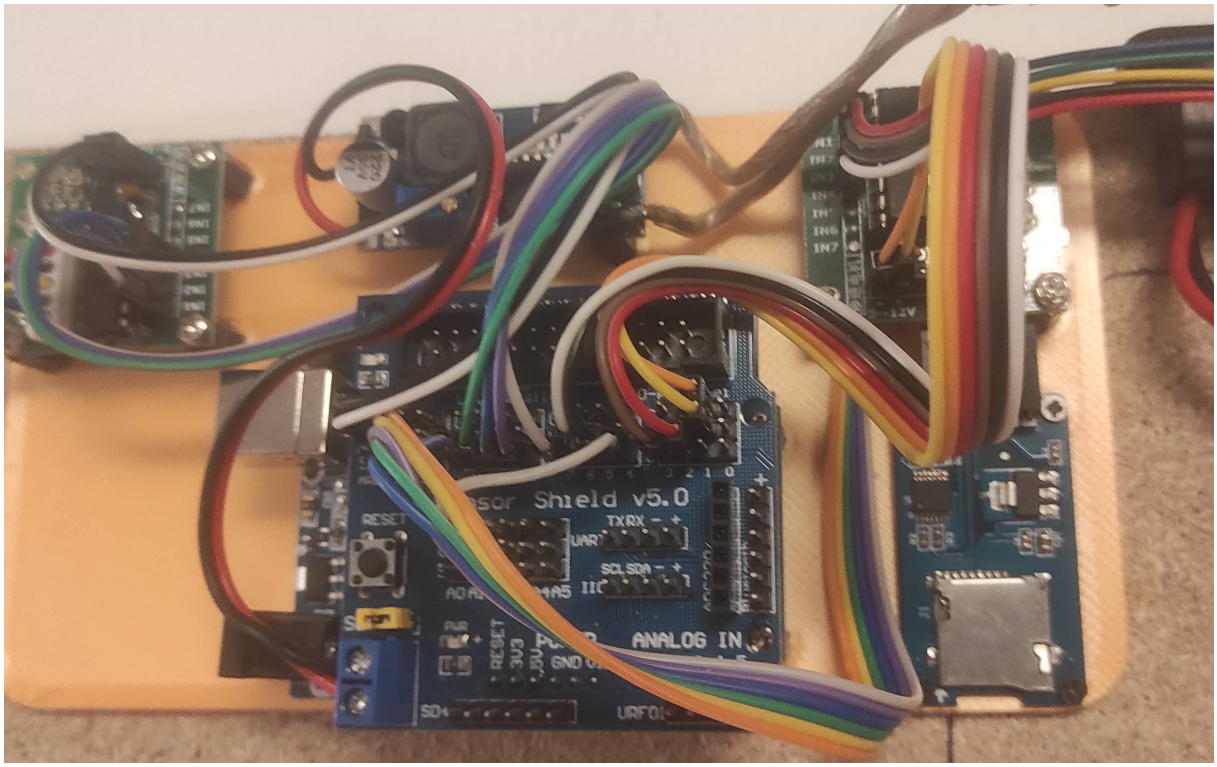
Such a plotter is available as a wooden kit on well-known Asian shopping sites, you can even find an “Arduino Polargraph” on eBay for €69. Somehow it's all only half useful. Even painting on a vertical surface is difficult with this. In any case, I have never seen a construction that you can simply clamp to the nearest whiteboard and let run. Everything can be done better and much cheaper. That's why I took on this kit and designed the parts for 3D printing.

The project with the UNO I found bad primitive and unwieldy, I developed a variant with an ESP32. However, I'm also showing the UNO version here in case someone doesn't like tinkering with electronics.

Plotter with Arduino UNO

Whom the UNO that's enough - here you go: printing 3D files, build UNO etc. by image, wire it, download the code and off you go. The Parts list, photos and code are available on GitHub under

https://github.com/Pontifex42/Walldraw

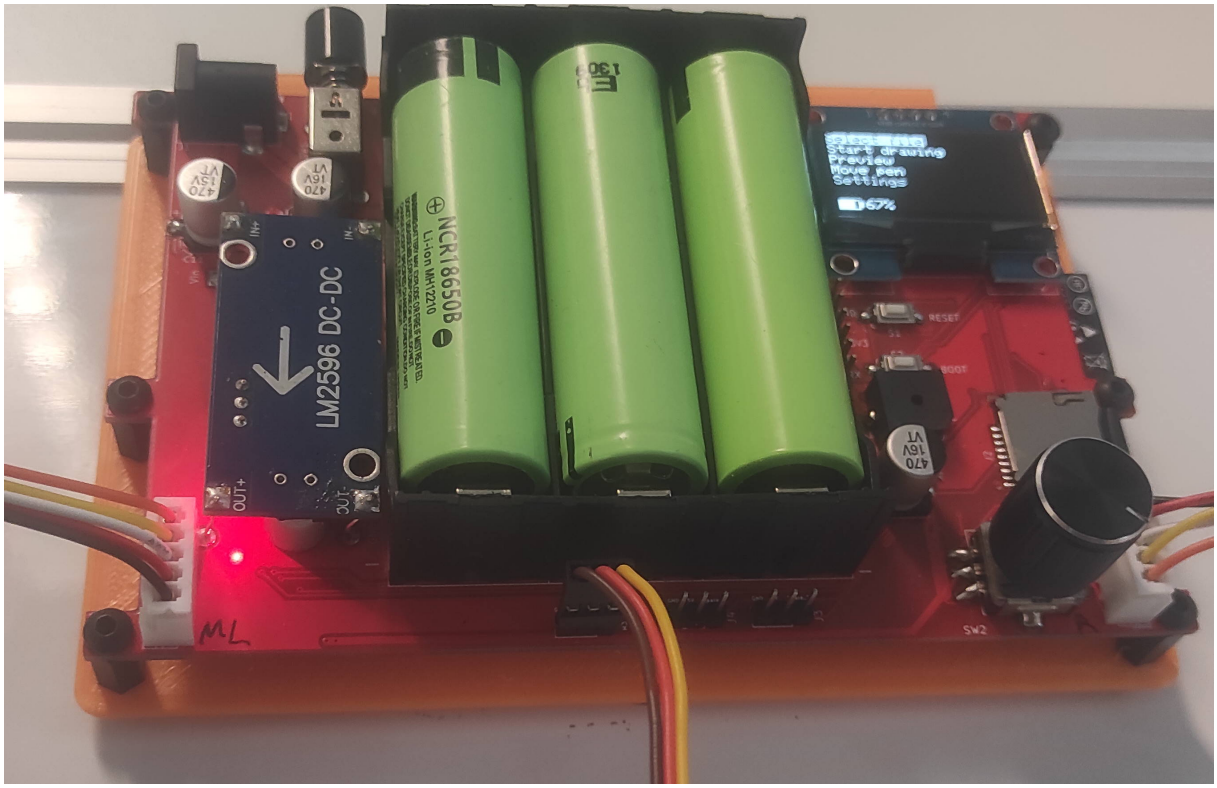


The disadvantages of the UNO variant:

1) I don't have any good instructions or circuit diagrams for this. But it shouldn't be a big problem, the pinout is in the code, and the rest should be clear from the photos.
2) A power supply is initially only possible with a 5V wall adapter. The power cable is then quite in the way. And if there is no power outlet nearby, there is no art prize to be won anyway. A power supply with a LiPo battery, for example, only requires a DC-DC converter and a few XT-60 Connecting plug, preferably anotherXT-60 switch. See photos.
3) The image must exist on the SD card as a file called “1.nc”, which is already created for the appropriate paper format scaled. If you want to draw a different picture or change the size, you carry the SD card first of all to the next PC. Anyway, to adapt an image NC file to a paper format, I wrote a DOS command-line tool called "WallDrawResizer.exe," which can also be found in the project. Otherwise, you take Inkscape and learn how to use it first.
4) The code must be adapted to e.g. the up/down angle of the servo (Important!), the size of the plotter (Default: 1000mm), the direction of rotation of the motors, the speed, and more. The whole thing is actually designed to be permanently mounted on a board, which you can see in the pictures.
5) After turning on the program, it starts drawing around the center of the drawing area. It can't do anything more.

Plotter with ESP32

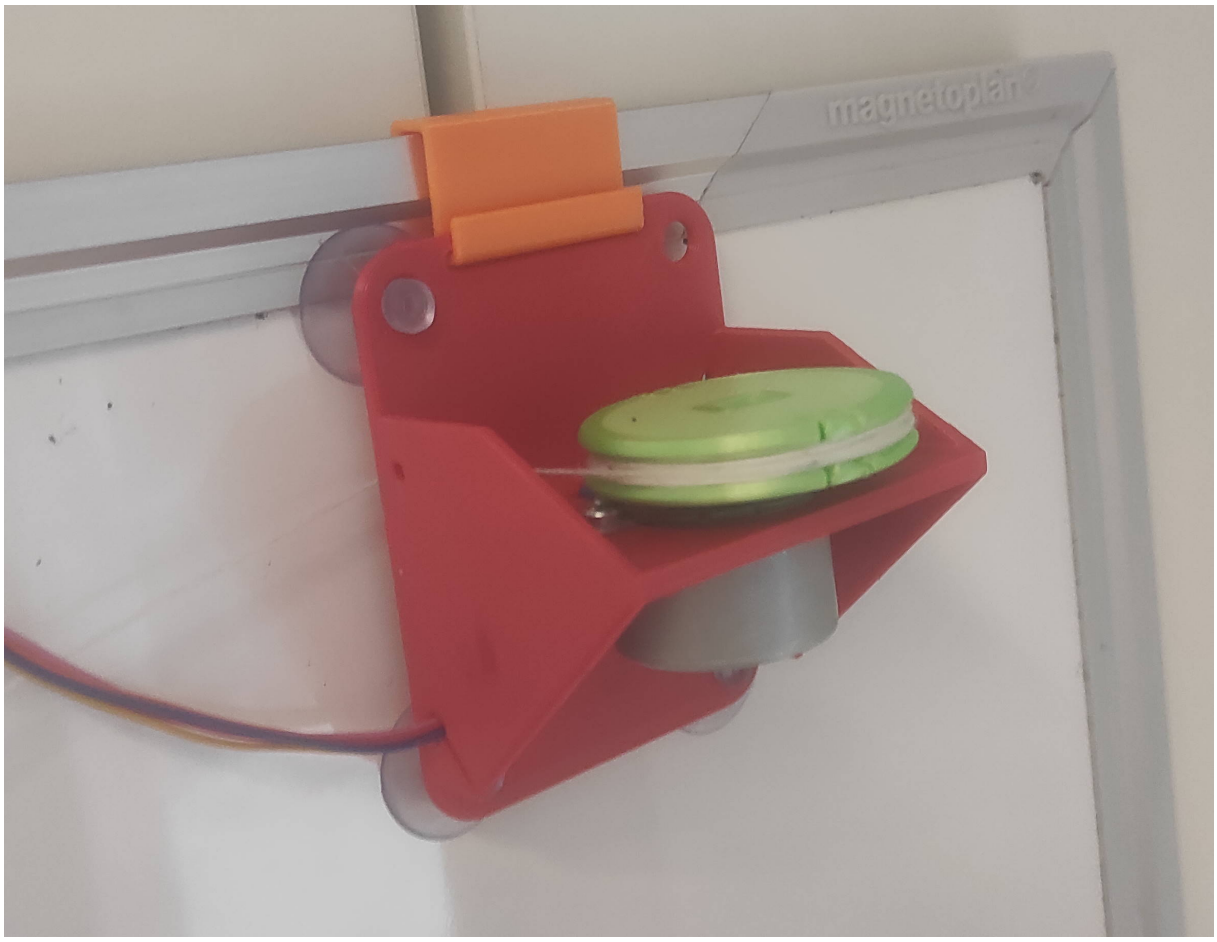
For those who like a little challenge or simply like to tinker with electronics, there is a homemade circuit board and a “fat” processor. An ESP32 has enough computing power to even convert an NC file to the desired paper format. A display allows you to select files, adjust settings flexibly instead of changing them in the code, move the drawing slightly in the drawing area, and much more. And most importantly, a couple of batteries make the whole thing usable anywhere: house walls, office whiteboards, you name it.



If you've always wanted to try having a circuit board made (you can order one for just over €6) and soldering SMD components, go for it! It's not that difficult if you watch a few YouTube tutorials beforehand. As an electronics enthusiast, you'll probably already have most of the parts in your drawer in the usual kits. However, it will be a bit more expensive if you don't have a halfway decent electronics workshop.

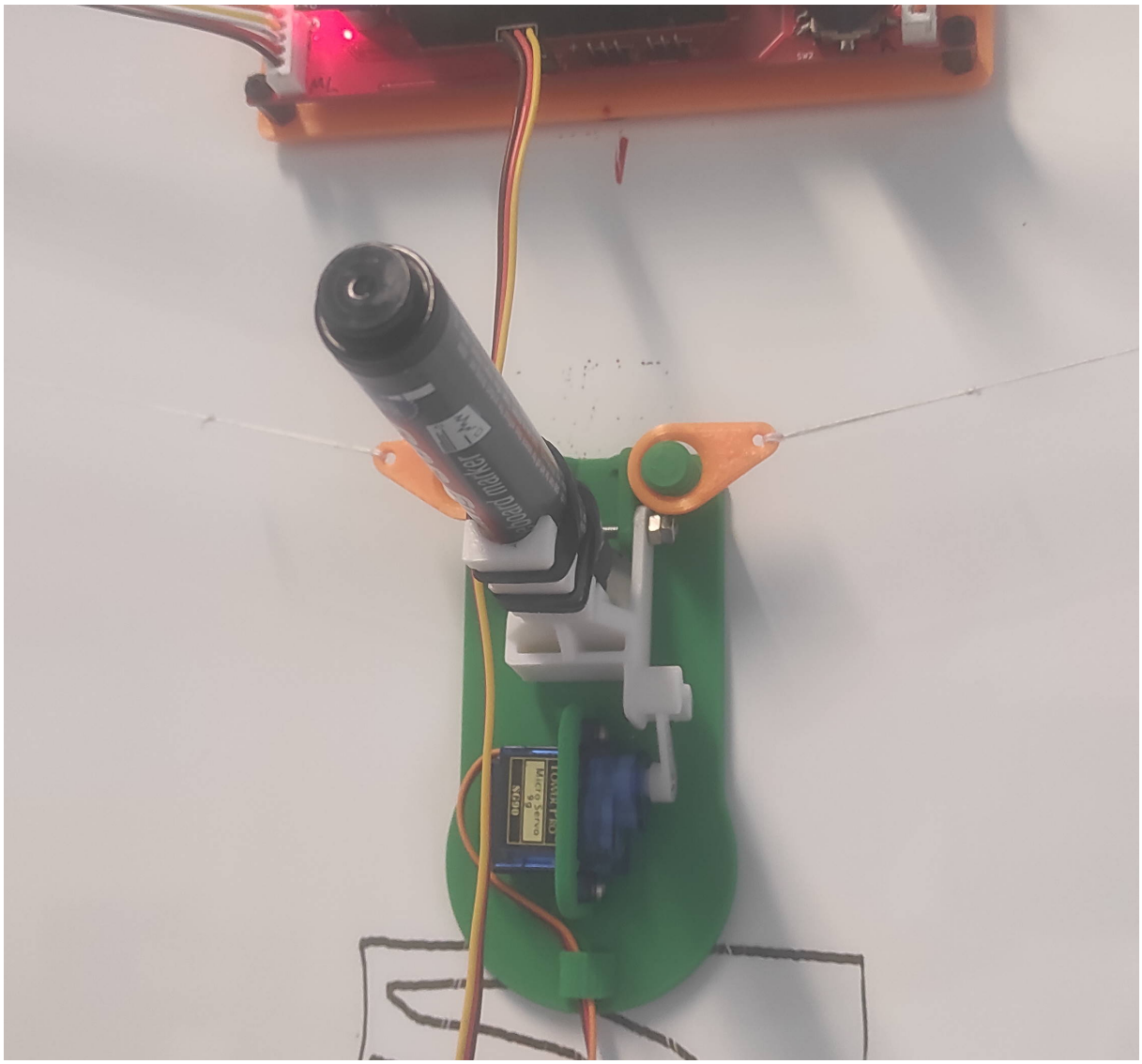
The 3D files and structure of the parts

The archive files for Fusion360 are included in the downloads, just in case someone wants to adjust or improve a design. Otherwise: put the STL files into the slicer and quickly start the printer. All STL files are easy to print. Support structures can be enabled for the stepper mounts, but they also work without them. Everything else requires neither a brim nor supports.



The hooks are designed to easily hang the pieces from the top edge of a whiteboard. They're intentionally a bit difficult to hook into place. Alternatively, you can screw the whole thing onto a board. The rubber suction cups can be inserted with pliers and a little force, a little practice helps. But you don't have to have it, some tape to prevent the motors from slipping will do the trick.

The holes for the motor axes are intentionally a bit small to ensure a tight fit. Depending on the printer's calibration, you may need to file them down a tiny bit. The same applies to the servo screws. The pen holder is attached to the carriage with M3 screws or an M3 threaded rod. If you don't have self-locking nuts, you should use lock nuts.



The strings can be secured in the notches of the spools for winding. Thread the other end of the string through one of the small holes, tie it into a loop, and attach it either to the string rings or directly to the carriage. The right and left motors naturally use the left and right holes. If you're not screwing the stepper mounts to a board and have an air gap at the back, you should use the holes on the underside; this makes balancing the pen easier later.

If cables are dangling, the cable clamps can help, but otherwise, they're not needed. You only print the version of the mounting plates you want to build.

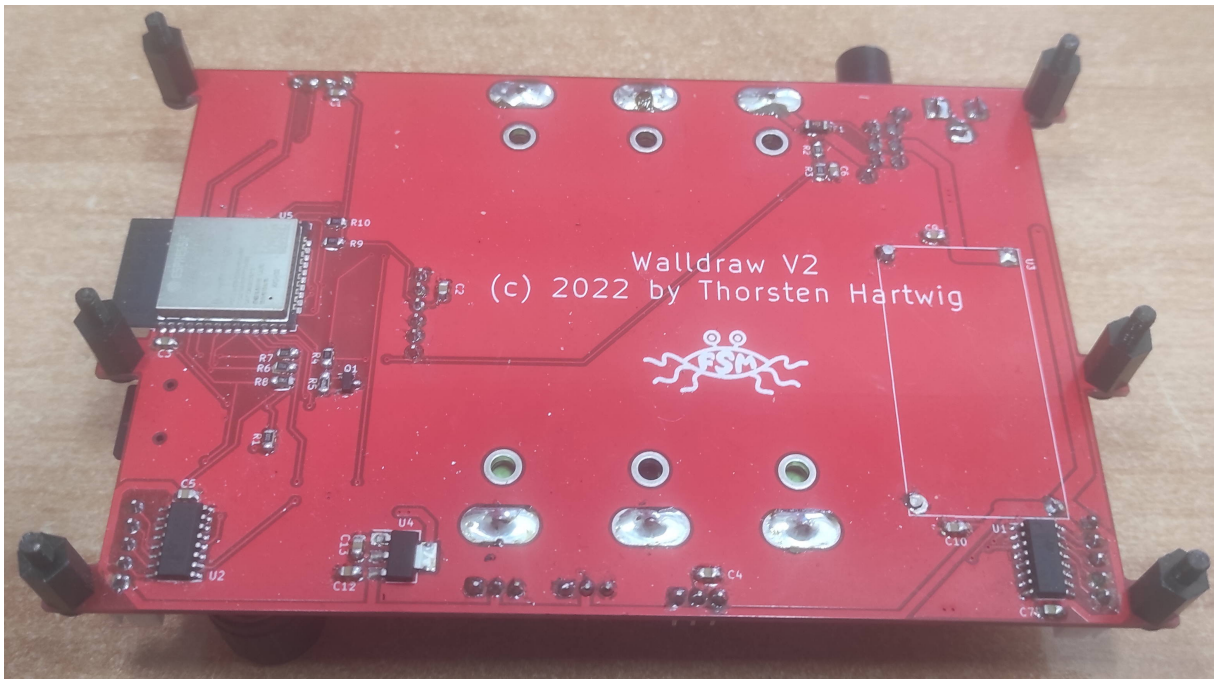
The rest should become clear from the pictures.

The board with the ESP32

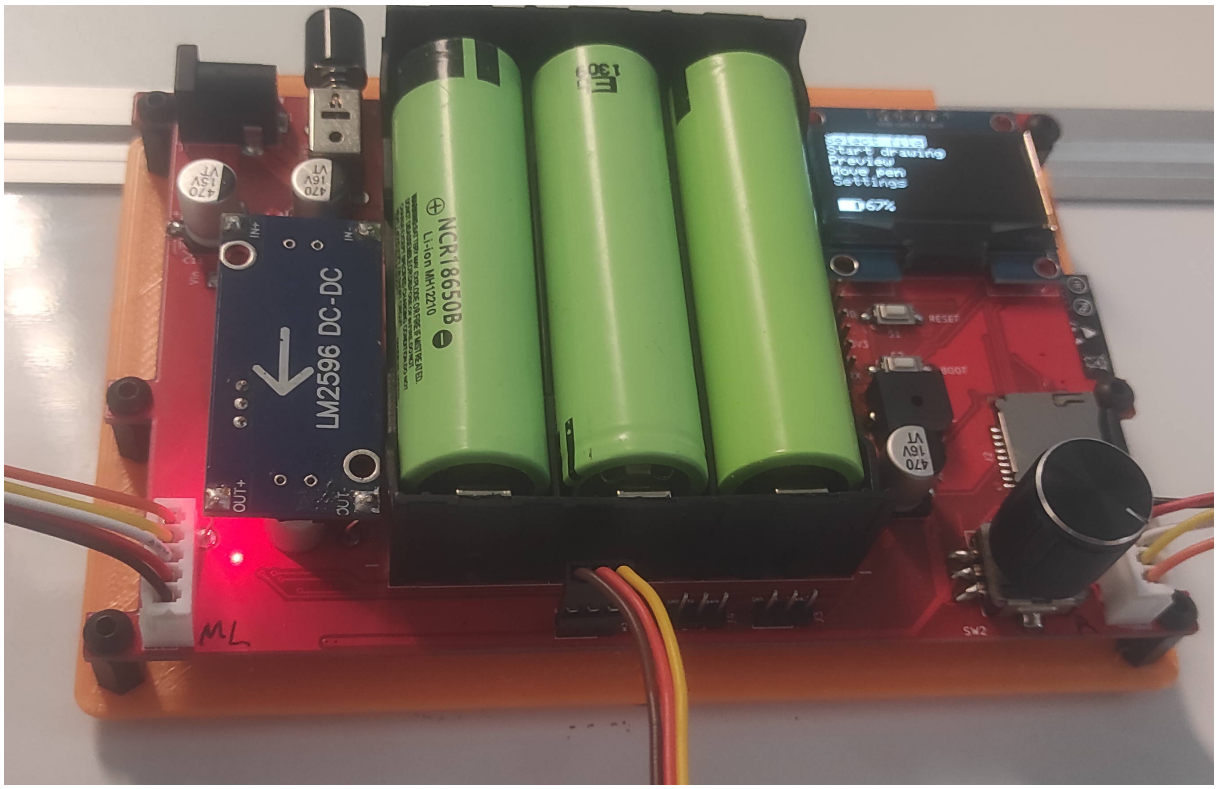
The Gerber files for manufacturing the circuit board are included in the project, but can also be created using KiCad. The zip file can be uploaded to sites like JLCPCB or PCBWAY and quickly clicked through the ordering process.The parts will be in your mailbox in about 12 days.

The file ibom.html contains the parts list with links to possible suppliers. I don't think I used anything complicated or unusual. If you have any questions, please contact me directly. Asking isn't a waste: walldraw (at) hartwig.org

First, you assemble the bottom panel. Soldering the ESP32 module may be a bit of a hurdle, but I apply solder paste with a stencil and place it on a hot plate. Otherwise, don't be stingy with the flux, and it'll work.



TheDC-DC converter is mounted upside down with wires to prevent anyone from turning the adjustment screw and damaging the entire thing. So don't forget to set the voltage to 5.0V beforehand.



Some components aren't absolutely necessary. The beeper is actually only needed for battery warning, the electrolytic capacitors are more for safety, it should work without them. The "spare" pins are superfluous. Some design the FTDI connector so that the FTDI module plugs directly into the holes. and you probably only press the buttons once to download the firmware, so a piece of wire is enough. Of course, the 5.5mm DC jack isn't needed if you're using a battery as the power source. The reverse applies to the battery holder. And who needs a backup anyway? You could also save the SD card reader if you program a web interface and view the images via browser/smartphone into the memory of the ESP32, there are at least 3MB free (there are also modules with 8/16MB). I didn't bother, this is whatmore dedicated hobbyists.

The software

To make the code easier for non-programmers to understand, I've omitted object orientation. I think the whole thing is somewhat readable thanks to the "software architecture" approach. The code can be compiled and loaded with the Arduino IDE 2.x. For the ESP variant you need the OLED-Lib Adafruit_SH110X (with Adafruit_GFX), for the UNO variant the lib TinyStepper. I personally use Microsoft Visual Studio with the plugin "VisualMicro Arduino," which is why it includes an .sln and .vcxproj file. The plugin costs $12 per year after the 45-day trial period for hobbyists and students.

That's all there is to it for the UNO. For ESP32, there are two more pitfalls: it Board Manager versions 1.0.6 and 2.x are currently in circulation. Please update to 2.x. Besides that, the Arduino IDE provides the lib “SD.h”. You have to remove it from the lib directory there, so that the “SD.h” lib of theESP32 is used – it's called the same thing there. Caution: If you set the FTDI adapter to 5V instead of 3.3V, you'll have to solder a new board. Congratulations.

To download, if you're not familiar with it: Connect the FTDI adapter, then press the "Boot" and "Reset" buttons, then release Reset, then Boot. The ESP32 will then enter download mode.

The facilityof the ESP32

Once the plotter is set up, first adjust the servo's up and down angles in the "Settings – Servo angles" menu. If the servo is forced into the down position against resistance, it will burn out after a short time. Therefore, leave some play in the down position.

Next, measure the distance between the two thread holes and enter this under "Settings - Plotter dimensions - Set plotter width." An estimate of the height of the drawing area in "Settings - Plotter dimensions - Set plotter height" is sufficient. Then, test the direction of rotation of the steppers under "Move pen - Move vertical." Negative Y values should go downwards. If it rotates the other way, either reverse the spools or select "Settings - Reverse Motors."

All settings remain saved after switching off.

Since the pen cannot be positioned horizontally between the thread holes, but must remain slightly below them, the upper edge of the drawing area is set at one-eighth of the width below the thread holes, i.e., for a 1500mm width, approximately 19 cm lower. Before starting to draw, the pen must be brought to this height by rotating the spools, in the middle between the thread holes—this is the starting or zero position. (On the UNO, this is the center of the drawing area.)

Inserting a pen correctly is a bit tricky. Sometimes it's too high and never draws, sometimes it's too low and always draws, so you have to experiment for a moment before hanging the slide on the threads. The notches serve to balance the slide; this depends on the weight of the pen.

The operation

Unless a further offset is entered, the top edge of the drawing or paper is approximately 1/6 of the plotter width below the top edge (on the UNO: in the center of the plotter area). With a 1000mm pitch between the filament holes, this is 17cm below the holes, horizontally in the middle between the holes. This is where the pen must initially be positioned by turning the spindles. Now you can use “Select file” to select a file from the SD card. Select it, display it on the display using "Preview," and then click "Start drawing.". A file called "Calibrate.nc" that draws a rectangle for testing is included in the project.

With "Speed" you can change the speed slightly. Level 10 could increase the speed of the engines except kick, levels 9 and 8 produce somethingi mprecise lines. The smaller stages are used to extend a drawing operation to the whole day for demonstration purposes.

One battery charge should be enough for more than five hours of painting. Be careful with deeply discharged batteries: if "Battery Power Low" is displayed, turn off the device as soon as possible. There is no processor shutdown feature implemented, only the motors are shut down. If the batteries are deeply discharged, they are damaged. The DC converter requires some power even when idle.

Under “Settings-Paper size” you can select a paper size. The drawing will be resized so that a 1cm margin remains. The "Native" setting draws at the size specified by the file. With "Settings-Draw offsets," you can position the drawing in the drawing area, for example, to create multiple drawings on one sheet.

Drawings

What's still missing are drawings. I'm a bit hesitant about publishing my drawings, as that might run into copyright issues. So, here's a quick guide to creating files yourself:

- Search for a nice drawing online. The best results are obtained with sketches without grayscale or color. For example, google "Einstein line drawing."or simply “drawing”
- Downloading and installing the software “Inkscape” is free.
- Watch a tutorial on YT, for example, with the search phrase “Create Inkscape gcode”
- In my implementation, the files must have the file extension “.nc”, just rename the .ngc files from Inkscape.
- For the UNO, resize the file to the desired size with “WallDrawResizer.exe” and rename it to “1.nc”
- Put it on the SD card and then play around with the plotter's menus.
- Let's go.

Did I forget something? Comments and emails are welcome. Improvements to the software anyway.

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